

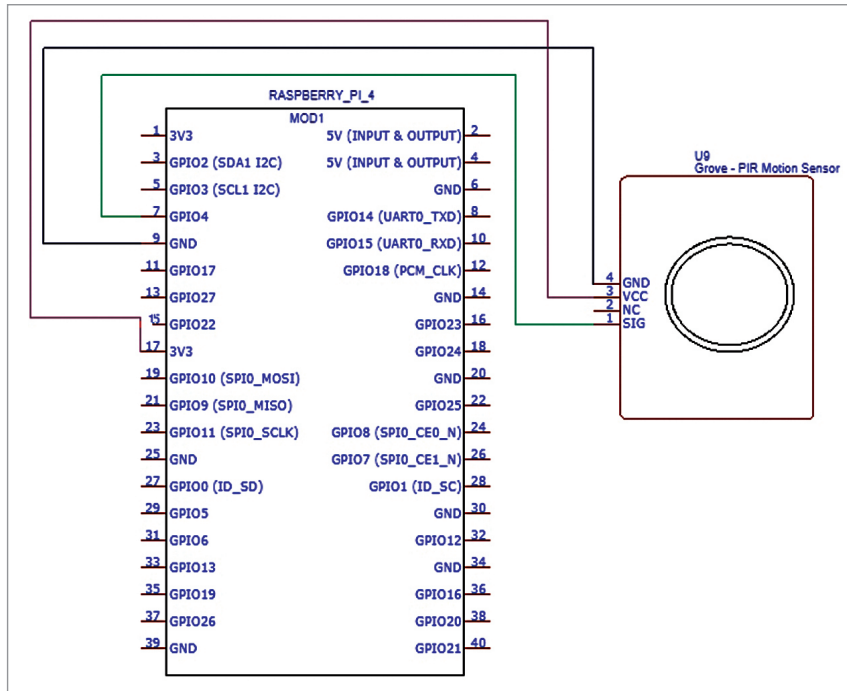


Fig. 3: Circuit diagram

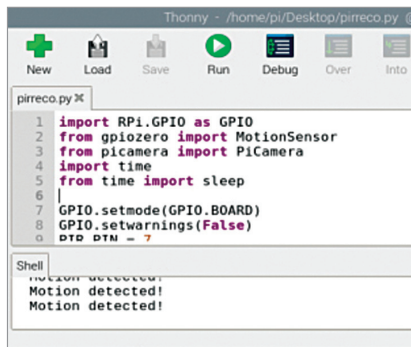


Fig. 4: Testing code of Pi camera

the prototype when the sensor detects a small hand movement and activates the Pi camera to record the image. The image is captured and stored for reference. The components used in the project are listed in Table 1 for the Bill of Material.

Installation and setup

To use a Raspberry Pi camera (having a ribbon cable), first enable the camera interface in Raspberry Pi config setting. The following code is used for this part:

```
$ sudo apt-get install python-picamera
$ sudo apt-get install python3-picamera
```

TABLE 2
PIN CONNECTIONS
BETWEEN PIR SENSOR
AND RASPBERRY PI

Board Label	Pin Function (PIR)	Raspberry Pi
GND	Ground	GND
5V	5V power input	5V
OUT	Output of the sensor	Pin 7 (GPIO 4)

Then follow the step by step instructions given below.

Step 1. Enable Raspberry Pi camera software before using the camera module. In the desktop environment, first go to Menu and select Preferences→Raspberry Pi Configuration. Select the Interfaces tab and Enable the camera.

Step 2. After successfully completing Step 1, update the software by running the following Linux commands in the Linux terminal:

```
$ sudo apt-get update
$ sudo apt-get upgrade
```

Step 3. After software update, reboot the system.

Step 4. After reboot, write the

code in Python3/ThonnyIDE for working of the project.

Circuit and testing

The circuit is quite simple, as shown in Fig. 3 for the circuit diagram.

Refer Table 2 for the connections between PIR sensor and the Raspberry Pi board.

Test the code of the Pi camera in terminal by running the command below for 5 seconds. This ensures that motion is detected. Refer Fig. 4 for testing code of the Pi camera in terminal.

```
raspistill -o test.jpg
```

As we know, memory storage requirement of each clicked image is large. Hence, store the images in Firebase, which is a set of hosting services for any type of application. It offers NoSQL, real-time hosting of databases, content, social authentication, and notifications, or services such as a real-time communication server.

Following are the stepwise instructions for new users:

1. Open the website <https://firebase.google.com>

A Firebase project is a container for your Apple, Android, and Web apps. Apps in this project will share features like Database, User Management, and Remote Configuration. A Firebase project is a cloud project. When you create a new Firebase project in Firebase console, you are actually creating a Google Cloud project behind the scenes.

2. Click on Get Started button, which will lead to the project console <https://console.firebase.google.com>

Click on Create a Project button and create a project. Give a name to the project; here the given project name is Intrusion Detection. Accept the terms of Firebase and press Continue button.

3. Configure Google Analytics, which is a free and unlimited analytics solution that enables targeting, reporting, and more in Firebase